

Muhamad Assem Ahmed

Lab 14

Part A

```
=====
Session 14 PreInit - VERIFY ONLY (nothing is changed)
=====
Database ITI_StudentPortal      : FOUND
Table Students                  : FOUND (4 row(s))
Table Courses                   : FOUND
Table Instructors               : FOUND
Migration history                : 2 migration(s) applied
Instructors.AssignedCourseName  : PRESENT (expected - Block 3 removes it)
Courses.InstructorId            : ABSENT (expected - Block 3 adds it)
=====
PreInit complete. Ready for Session 14.
Nothing was created, dropped, or altered by this script.
=====

Completion time: 2026-07-29T12:40:33.6892416+03:00
```

Part B

B1 : Prints 3.99. but nothing has been sent to the database yet

B2: prints 0 for every instructor because ToListAsync() loads only the Instructor table
It does not load related Courses

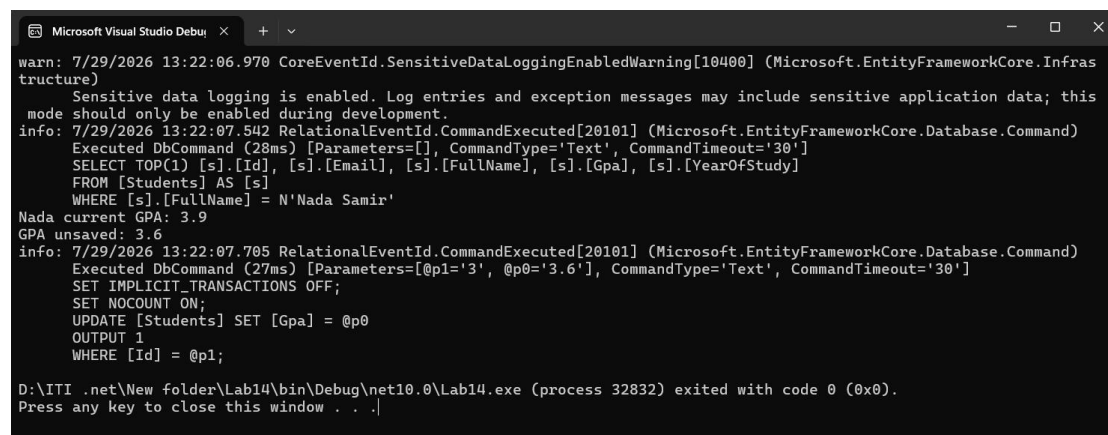
B3 : Nothing is saved to the database because AsNoTracking() tells Entity Framework Core not to track the entity it retrieves only modifies the object in memory

Part C

GPA = $3.0 + ((27 \% 7) * 0.1) = 3.0 + (6 * 0.1) = 3.6$

C2: still show the old value: SaveChangesAsync() hasnot run

C3:



```
Microsoft Visual Studio Debug Console
warn: 7/29/2026 13:22:06.970 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infras
structure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this
mode should only be enabled during development.
info: 7/29/2026 13:22:07.542 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (28ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT TOP(1) [s].[Id], [s].[Email], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
FROM [Students] AS [s]
WHERE [s].[FullName] = N'Nada Samir'
Nada current GPA: 3.9
GPA unsaved: 3.6
info: 7/29/2026 13:22:07.705 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (27ms) [Parameters=[@p1='3', @p0='3.6'], CommandType='Text', CommandTimeout='30']
SET IMPLICIT_TRANSACTIONS OFF;
SET NOCOUNT ON;
UPDATE [Students] SET [Gpa] = @p0
OUTPUT 1
WHERE [Id] = @p1;
D:\ITI .net\New folder\Lab14\bin\Debug\net10.0\Lab14.exe (process 32832) exited with code 0 (0x0).
Press any key to close this window . . .|
```

C4:

```
Microsoft Visual Studio Debug
warn: 7/29/2026 13:27:49.399 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infras
tructure)
    Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this
    mode should only be enabled during development.
info: 7/29/2026 13:27:50.000 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
    Executed DbCommand (28ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
    SELECT TOP(1) [s].[Id], [s].[Email], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
    FROM [Students] AS [s]
    WHERE [s].[FullName] = N'Nada Samir'
Nada current GPA: 3.6
GPA unsaved: 3.6
info: 7/29/2026 13:27:50.168 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
    Executed DbCommand (28ms) [Parameters=[@p0=NULL (Size = 4000), @p1='Muhamad Assem Ahmed' (Nullable = false) (Size
    = 4000), @p2='3.6', @p3='2'], CommandType='Text', CommandTimeout='30']
    SET IMPLICIT_TRANSACTIONS OFF;
    SET NOCOUNT ON;
    INSERT INTO [Students] ([Email], [FullName], [Gpa], [YearOfStudy])
    OUTPUT INSERTED.[Id]
    VALUES (@p0, @p1, @p2, @p3);
Id after save: 5
D:\ITI .net\New folder\Lab14\bin\Debug\net10.0\Lab14.exe (process 35196) exited with code 0 (0x0).
Press any key to close this window . . .|
```

C5:

```
Microsoft Visual Studio Debug
Id before save: 0
warn: 7/29/2026 13:33:55.075 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infras
tructure)
    Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this
    mode should only be enabled during development.
info: 7/29/2026 13:33:55.600 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
    Executed DbCommand (37ms) [Parameters=[@p0=NULL (Size = 4000), @p1='Muhamad Assem Ahmed' (Nullable = false) (Size
    = 4000), @p2='3.6', @p3='2'], CommandType='Text', CommandTimeout='30']
    SET IMPLICIT_TRANSACTIONS OFF;
    SET NOCOUNT ON;
    INSERT INTO [Students] ([Email], [FullName], [Gpa], [YearOfStudy])
    OUTPUT INSERTED.[Id]
    VALUES (@p0, @p1, @p2, @p3);
Id after save: 7
info: 7/29/2026 13:33:55.655 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
    Executed DbCommand (11ms) [Parameters=[@p1='7', @p0='3'], CommandType='Text', CommandTimeout='30']
    SET IMPLICIT_TRANSACTIONS OFF;
    SET NOCOUNT ON;
    UPDATE [Students] SET [YearOfStudy] = @p0
    OUTPUT 1
    WHERE [Id] = @p1;
year of study after update: 3
D:\ITI .net\New folder\Lab14\bin\Debug\net10.0\Lab14.exe (process 11112) exited with code 0 (0x0).
Press any key to close this window . . .|
```

Part D

What operation does Up perform on FullName?

Up performs: AlterColumn<string>

What are nullable: and oldNullable: set to, and what does the difference mean?

nullable: false, old nullable: false→ the column was not nullable this migration makes it not null but but in real it should change it from nullble to not nullable

What two kinds of existing row could make this migration fail?

rows with FullName is null, or rows where LEN(FullName) > 100

```
SELECT * FROM Students WHERE FullName IS NULL OR LEN(FullName) > 100
```

Results				
Id	FullName	YearOfStudy	Gpa	Email

Part E

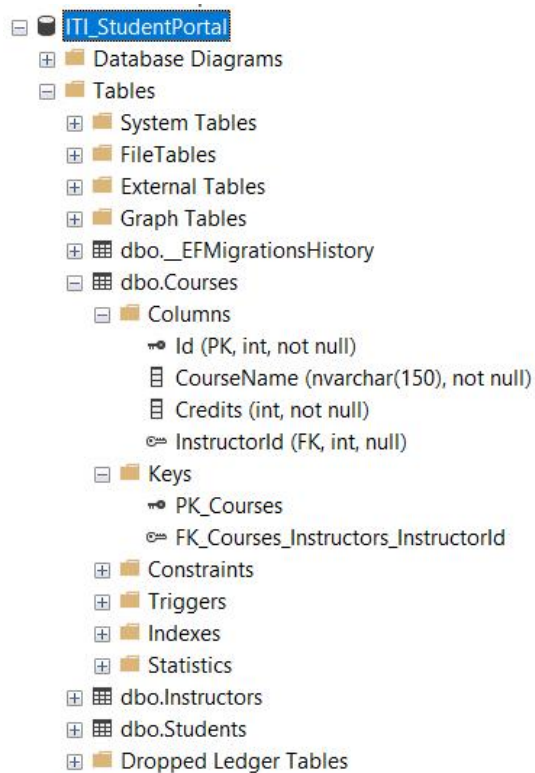
keeping both would be a mistake: two competing sources of truth for the same fact

List every operation Up performs.?

Up performs: AddColumn InstructorId DropColumn AssignedCourseName AddForeignKey CreateIndex

One of them destroys data. Which one, and what data?

DropColumn AssignedCourseName



```

SELECT TOP (1000) [Id]
      ,[CourseName]
      ,[Credits]
      ,[InstructorId]
FROM [ITI_StudentPortal].[dbo].[Courses]

```

Results			
Id	CourseName	Credits	InstructorId
1	Web Development Using .NET	5	1
2	Database Fundamentals	5	NULL

Part F

Load all instructors without Include, and loop printing each one's name and Courses.Count. Record: the counts printed, and how many SQL queries the log shows.
every count prints 0

Load them again with Include, loop with a nested inner loop printing course names. Record the counts and the query count from the log.
real counts print, and the log shows exactly 1 query

In a comment, answer: the Include version returned more rows from SQL Server than there are instructors. Explain why, and what EF did with the duplicates.
Include's sql returns more rows than there are instructors because the JOIN produces one row per (Instructor, Course)

Part G

Part C GPA = $3.0 + ((27 \% 7) * 0.1) = 3.6$

Part E OnDelete = $27 \% 2 = 1 \rightarrow \text{SetNull(InstructorId} \rightarrow \text{int ?)}$

Part F extra courses = $(27 \% 3) + 2 = 2$

OnDelete = SetNull requires the FK to be nullable

no migration failed

both come from re-hitting the database every time you touch data multiple enumeration re runs the whole linq N+1 re-runs in each row query each time you touch a related